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Java SE 27 & JDK 27

Class FileHandler

java.lang.Object
    java.util.logging.Handler
        java.util.logging.StreamHandler
            java.util.logging.FileHandler

```
public class FileHandler
extends StreamHandler
```

Simple file logging `Handler`.

The `FileHandler` can either write to a specified file, or it can write to a rotating set of files.

For a rotating set of files, as each file reaches a given size limit, it is closed, rotated out, and a new file opened. Successively older files are named by adding "0", "1", "2", etc. into the base filename.

By default buffering is enabled in the IO libraries but each log record is flushed out when it is complete.

By default the `XMLFormatter` class is used for formatting.

Configuration: By default each `FileHandler` is initialized using the following `LogManager` configuration properties where `<handler-name>` refers to the fully-qualified class name of the handler. If properties are not defined (or have invalid values) then the specified default values are used.

- <handler-name>.level specifies the default level for the `Handler` (defaults to `Level.ALL`).
- <handler-name>.filter specifies the name of a `Filter` class to use (defaults to no `Filter`).
- <handler-name>.formatter specifies the name of a `Formatter` class to use (defaults to `java.util.logging.XMLFormatter`)
- <handler-name>.encoding the name of the character set encoding to use (defaults to the default platform encoding).
- <handler-name>.limit specifies an approximate maximum amount to write (in bytes) to any one file. If this is zero, then there is no limit. (Defaults to no limit).
- <handler-name>.count specifies how many output files to cycle through (defaults to 1).
- <handler-name>.pattern specifies a pattern for generating the output file name. See below for details. (Defaults to "%h/java%u.log").
- <handler-name>.append specifies whether the FileHandler should append onto any existing files (defaults to false).
- <handler-name>.maxLocks specifies the maximum number of concurrent locks held by FileHandler (defaults to 100).

For example, the properties for `FileHandler` would be:

- java.util.logging.FileHandler.level=INFO
- java.util.logging.FileHandler.formatter=java.util.logging.SimpleFormatter

For a custom handler, e.g. com.foo.MyHandler, the properties would be:

- com.foo.MyHandler.level=INFO
- com.foo.MyHandler.formatter=java.util.logging.SimpleFormatter

A pattern consists of a string that includes the following special components that will be replaced at runtime:

- "/" the local pathname separator
- "%t" the system temporary directory
- "%h" the value of the "user.home" system property
- "%g" the generation number to distinguish rotated logs
- "%u" a unique number to resolve conflicts
- "%%" translates to a single percent sign "%"

If no "%g" field has been specified and the file count is greater than one, then the generation number will be added to the end of the generated filename, after a dot.

Thus for example a pattern of "%t/java%g.log" with a count of 2 would typically cause log files to be written on Solaris to /var/tmp/java0.log and /var/tmp/java1.log whereas on Windows 95 they would be typically written to C:\TEMP\java0.log and C:\TEMP\java1.log

Generation numbers follow the sequence 0, 1, 2, etc.

Normally the "%u" unique field is set to 0. However, if the `FileHandler` tries to open the filename and finds the file is currently in use by another process it will increment the unique number field and try again. This will be repeated until `FileHandler` finds a file name that is not currently in use. If there is a conflict and no "%u" field has been specified, it will be added at the end of the filename after a dot. (This will be after any automatically added generation number.)

Thus if three processes were all trying to log to fred%u.%g.txt then they might end up using fred0.0.txt, fred1.0.txt, fred2.0.txt as the first file in their rotating sequences.

Note that the use of unique ids to avoid conflicts is only guaranteed to work reliably when using a local disk file system.

Since:

1.4

Constructor Summary

Constructors

Constructor	Description
`FileHandler()`	Construct a default `FileHandler`.
`FileHandler(String pattern)`	Initialize a `FileHandler` to write to the given filename.
`FileHandler(String pattern, boolean append)`	Initialize a `FileHandler` to write to the given filename, with optional append.
`FileHandler(String pattern, int limit, int count)`	Initialize a `FileHandler` to write to a set of files.
`FileHandler(String pattern, int limit, int count, boolean append)`	Initialize a `FileHandler` to write to a set of files with optional append.
`FileHandler(String pattern, long limit, int count, boolean append)`	Initialize a `FileHandler` to write to a set of files with optional append.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description
`void`	`close()`	Close all the files.
`void`	`publish(LogRecord record)`	Format and publish a `LogRecord`.

Methods declared in class StreamHandler

`flush`, `isLoggable`, `setEncoding`, `setOutputStream`

Methods declared in class Handler

`getEncoding`, `getErrorManager`, `getFilter`, `getFormatter`, `getLevel`, `reportError`, `setErrorManager`, `setFilter`, `setFormatter`, `setLevel`

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Constructor Details

FileHandler

```
public FileHandler()
            throws IOException
```

Construct a default `FileHandler`. This will be configured entirely from `LogManager` properties (or their default values).

Throws:

`IOException` - if there are IO problems opening the files.

`NullPointerException` - if pattern property is an empty String.

FileHandler

```
public FileHandler(String pattern)
            throws IOException
```

Initialize a `FileHandler` to write to the given filename.

The `FileHandler` is configured based on `LogManager` properties (or their default values) except that the given pattern argument is used as the filename pattern, the file limit is set to no limit, and the file count is set to one.

There is no limit on the amount of data that may be written, so use this with care.

Parameters:

`pattern` - the name of the output file

Throws:

`IOException` - if there are IO problems opening the files.

`IllegalArgumentException` - if pattern is an empty string

FileHandler

```
public FileHandler(String pattern,
                   boolean append)
            throws IOException
```

Initialize a `FileHandler` to write to the given filename, with optional append.

The `FileHandler` is configured based on `LogManager` properties (or their default values) except that the given pattern argument is used as the filename pattern, the file limit is set to no limit, the file count is set to one, and the append mode is set to the given `append` argument.

There is no limit on the amount of data that may be written, so use this with care.

Parameters:

`pattern` - the name of the output file

`append` - specifies append mode

Throws:

`IOException` - if there are IO problems opening the files.

`IllegalArgumentException` - if pattern is an empty string

FileHandler

```
public FileHandler(String pattern,
                   int limit,
                   int count)
            throws IOException
```

Initialize a `FileHandler` to write to a set of files. When (approximately) the given limit has been written to one file, another file will be opened. The output will cycle through a set of count files.

The `FileHandler` is configured based on `LogManager` properties (or their default values) except that the given pattern argument is used as the filename pattern, the file limit is set to the limit argument, and the file count is set to the given count argument.

The count must be at least 1.

Parameters:

`pattern` - the pattern for naming the output file

`limit` - the maximum number of bytes to write to any one file

`count` - the number of files to use

Throws:

`IOException` - if there are IO problems opening the files.

`IllegalArgumentException` - if `limit < 0`, or `count < 1`.

`IllegalArgumentException` - if pattern is an empty string

FileHandler

```
public FileHandler(String pattern,
                   int limit,
                   int count,
                   boolean append)
            throws IOException
```

Initialize a `FileHandler` to write to a set of files with optional append. When (approximately) the given limit has been written to one file, another file will be opened. The output will cycle through a set of count files.

The `FileHandler` is configured based on `LogManager` properties (or their default values) except that the given pattern argument is used as the filename pattern, the file limit is set to the limit argument, and the file count is set to the given count argument, and the append mode is set to the given `append` argument.

The count must be at least 1.

Parameters:

`pattern` - the pattern for naming the output file

`limit` - the maximum number of bytes to write to any one file

`count` - the number of files to use

`append` - specifies append mode

Throws:

`IOException` - if there are IO problems opening the files.

`IllegalArgumentException` - if `limit < 0`, or `count < 1`.

`IllegalArgumentException` - if pattern is an empty string

FileHandler

```
public FileHandler(String pattern,
                   long limit,
                   int count,
                   boolean append)
            throws IOException
```

Initialize a `FileHandler` to write to a set of files with optional append. When (approximately) the given limit has been written to one file, another file will be opened. The output will cycle through a set of count files.

The `FileHandler` is configured based on `LogManager` properties (or their default values) except that the given pattern argument is used as the filename pattern, the file limit is set to the limit argument, and the file count is set to

the given count argument, and the append mode is set to the given append argument.

The count must be at least 1.

Parameters:

`pattern` - the pattern for naming the output file

`limit` - the maximum number of bytes to write to any one file

`count` - the number of files to use

`append` - specifies append mode

Throws:

`IOException` - if there are IO problems opening the files.

`IllegalArgumentException` - if `limit < 0`, or `count < 1`.

`IllegalArgumentException` - if pattern is an empty string

Since:

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Method Details

publish

`public void publish(LogRecord record)`

Format and publish a `LogRecord`.

Overrides:

`publish` in class `StreamHandler`

Parameters:

`record` - description of the log event. A null record is silently ignored and is not published

close

`public void close()`

Close all the files.

Overrides:

`close` in class `StreamHandler`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of

terms, workarounds, and working code examples. Other versions.
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